

Circular Disk Partition



Given a circular disk. You have to partition it in equal k pieces and all pieces should be of same shape. Find minimum cuts needed.

Input Format

First line contains T - number of test cases.

Next T lines containing an integer k - number of partitions for each test case.

Constraints

$$1 \leq T \leq 100$$

$$1 \leq k \leq 10^8$$

Output Format

Print T lines each containing an integer - number of minimum cuts needed.

Sample Input 0

```
1
4
```

Sample Output 0

```
2
```

Explanation 0

To cut the disk into 4 equal quarters, you have to make two cuts through the center with angle 90 degree between them.

Sample Input 1

```
2
4
5
```

Sample Output 1

```
2
5
```

Explanation 1

First testcase same as sample 0. For second testcase, to cut the disk into five equal parts you have to make five cuts.